CLOUD COMPUTING

LAB 7

MinIO and Ceph

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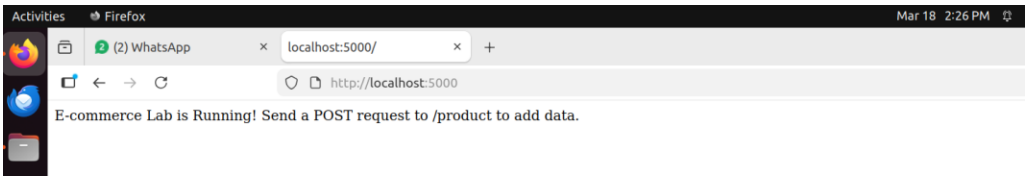
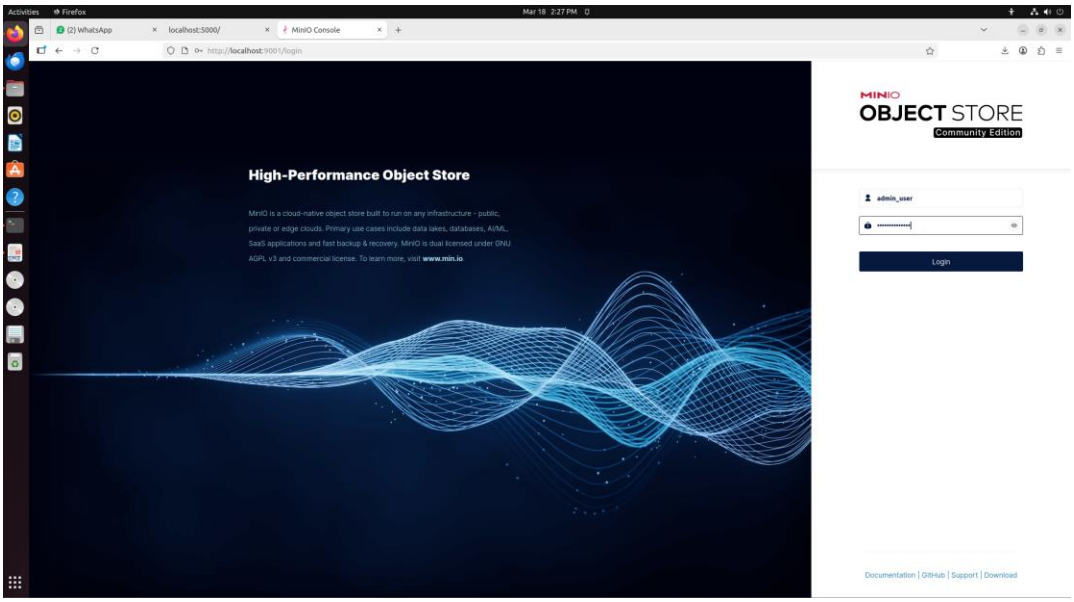
SRN: PES2UG23CS100

SECTION: B

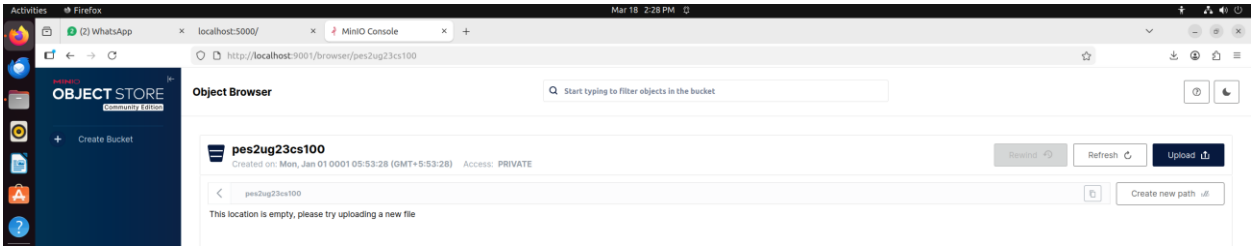
SS1:



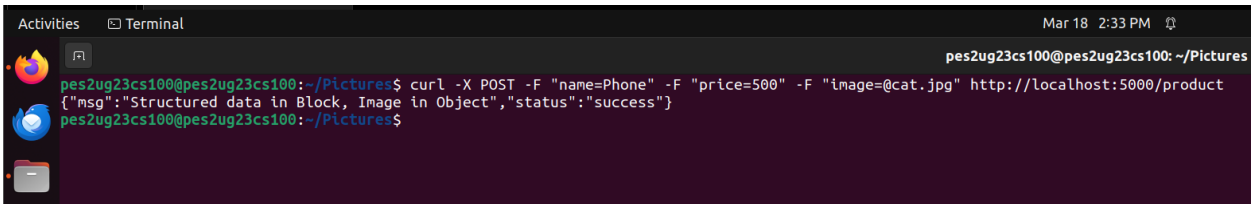
SS2:



SS3:

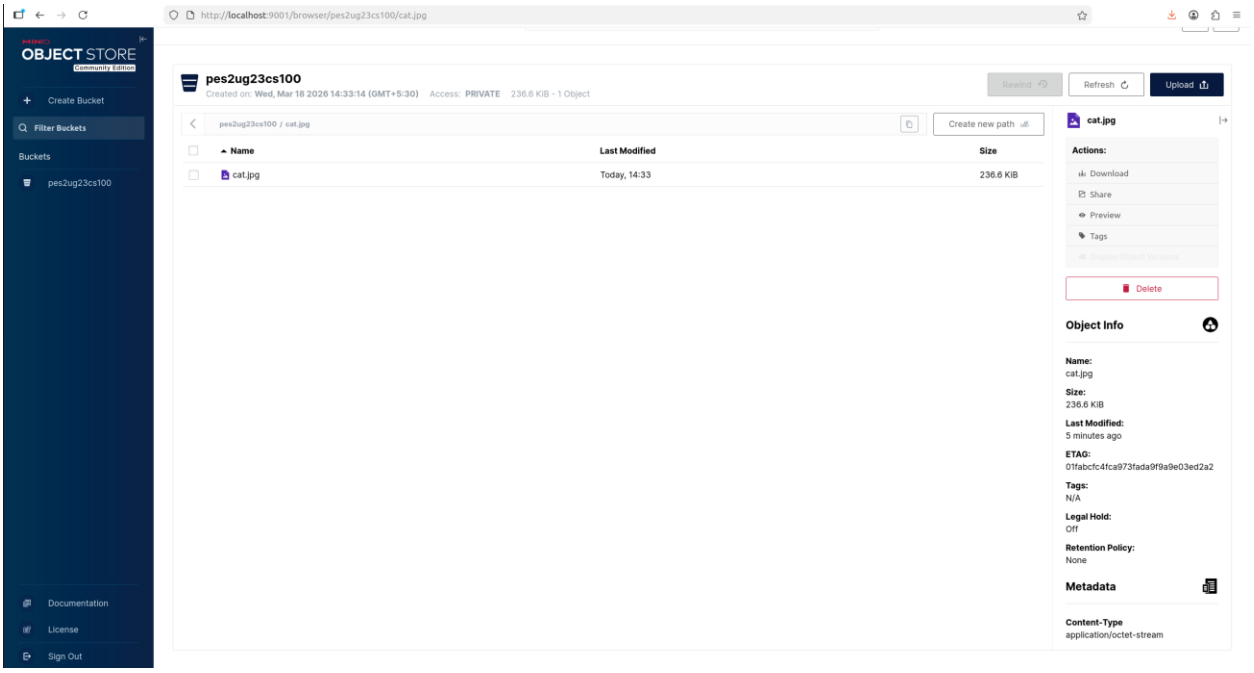


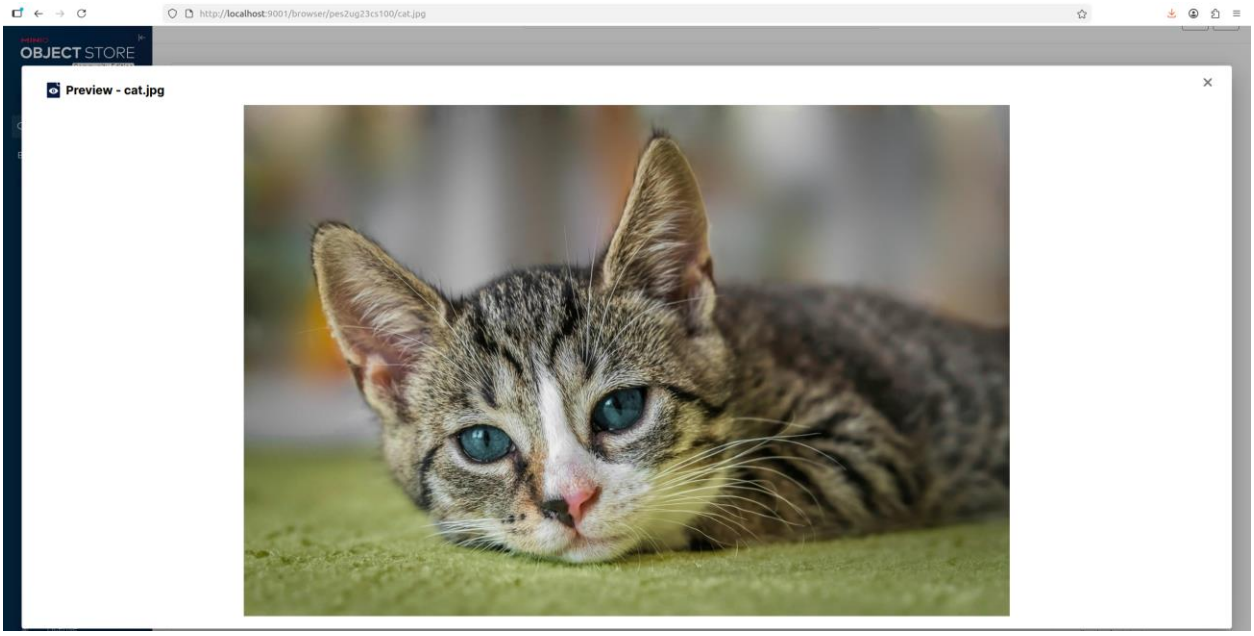
SS4:



Experiment 1:

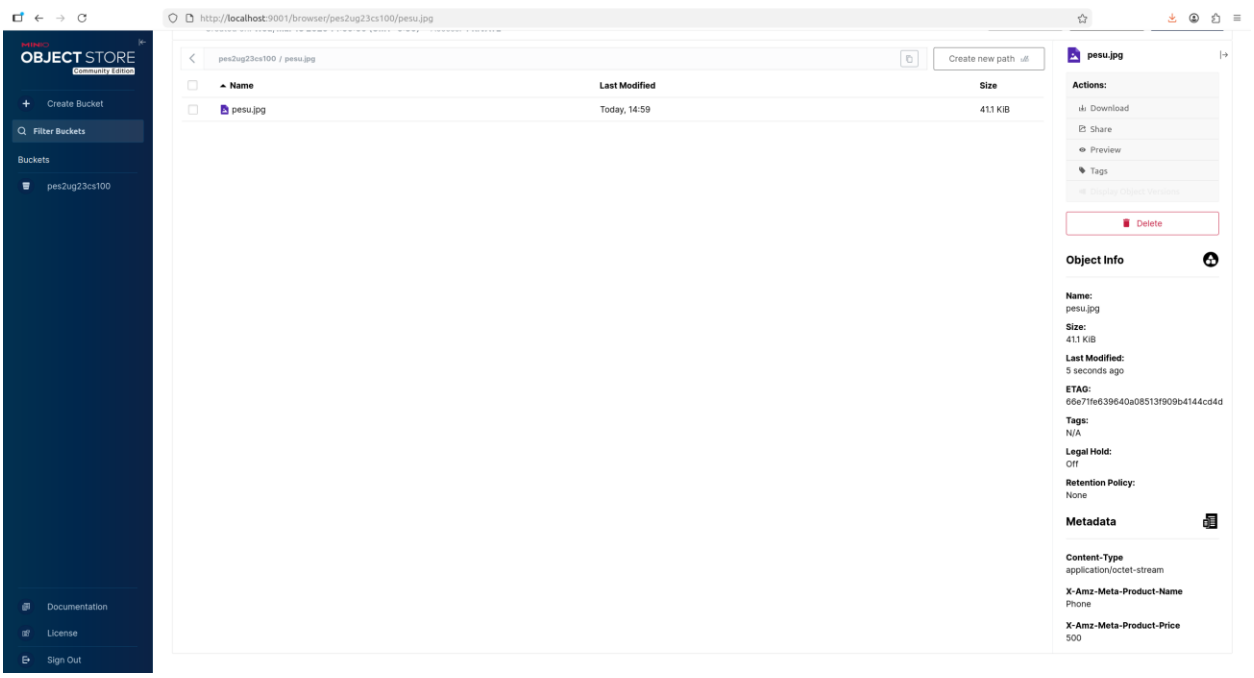
```
pes2ug23cs100@pes2ug23cs100:~/Pictures$ docker exec -it pes2ug23cs100-webapp-1 python -c "import sqlite3; conn =
sqlite3.connect('/mnt/block_volume/ecommerce.db'); cursor = conn.cursor(); cursor.execute('SELECT * FROM product'
); print(cursor.fetchall()); conn.close()"
[(1, 'Phone', 500.0, 'cat.jpg')]
pes2ug23cs100@pes2ug23cs100:~/Pictures$
```





Experiment 2:

```
pes2ug23cs100@pes2ug23cs100:~/Pictures$ curl -X POST -F "name=Phone" -F "price=500" -F "image=@pesu.jpg" http://localhost:5000/product
{"msg":"Structured data in Block, Image in Object","status":"success"}
pes2ug23cs100@pes2ug23cs100:~/Pictures$
```




```
pes2ug23cs100@pes2ug23cs100:~/Pictures$ docker exec -it pes2ug23cs100-webapp-1 python -c "import sqlite3; conn = sqlite3.connect('/mnt/
block_volume/ecommerce.db'); cursor = conn.cursor(); cursor.execute('SELECT * FROM product'); print(cursor.fetchall()); conn.close()"
[(1, 'Phone', 500.0, 'cat.jpg'), (2, 'Phone', 500.0, 'pesu.jpg')]
pes2ug23cs100@pes2ug23cs100:~/Pictures$
```

Experiment 3 (storage capacity limits):

```
pes2ug23cs100@pes2ug23cs100:~/Pictures$ curl -X POST -F "name=Phone" -F "price=500" -F "image=@image.jpg" http://localhost:5000/product
<!doctype html>
<html lang=en>
<title>500 Internal Server Error</title>
<h1>Internal Server Error</h1>
<p>The server encountered an internal error and was unable to complete your request. Either the server is overloaded or there is an error in the applicatio
```

```
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ docker-compose logs webapp
WARN[0000] /home/pes2ug23cs100/CC Lab/Lab7/PES2UG23CS100/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
webapp-1  | 🚀 Flask app running at http://localhost:5000
webapp-1  |  * Serving Flask app 'app'
webapp-1  |  * Debug mode: off
webapp-1  | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
webapp-1  |  * Running on all addresses (0.0.0.0)
webapp-1  |  * Running on http://127.0.0.1:5000
webapp-1  |  * Running on http://172.19.0.3:5000
webapp-1  | Press CTRL+C to quit
webapp-1  | 172.19.0.1 - - [18/Mar/2026 09:29:09] "POST /product HTTP/1.1" 200 -
webapp-1  | 172.19.0.1 - - [18/Mar/2026 09:41:25] "POST /product HTTP/1.1" 200 -
webapp-1  | 172.19.0.1 - - [18/Mar/2026 09:46:55] "POST /product HTTP/1.1" 200 -
webapp-1  | [2026-03-18 09:50:38,561] ERROR in app: Exception on /product [POST]
webapp-1  | Traceback (most recent call last):
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 1511, in wsgi_app
webapp-1  |     response = self.full_dispatch_request()
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 919, in full_dispatch_request
webapp-1  |     rv = self.handle_user_exception(e)
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 917, in full_dispatch_request
webapp-1  |     rv = self.dispatch_request()
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 902, in dispatch_request
webapp-1  |     return self.ensure_sync(self.view_functions[rule.endpoint])(**view_args)  # type: ignore[no-any-return]
webapp-1  |   File "/app/app.py", line 77, in add_product
webapp-1  |     minio_client.fput_object(
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/minio/api.py", line 1054, in fput_object
webapp-1  |     return self.put_object(
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/minio/api.py", line 2015, in put_object
webapp-1  |     raise exc
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/minio/api.py", line 1960, in put_object
webapp-1  |     return self._put_object(
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/minio/api.py", line 1789, in _put_object
webapp-1  |     response = self._execute(
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/minio/api.py", line 444, in _execute
webapp-1  |     return self._url_open(
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/minio/api.py", line 427, in _url_open
webapp-1  |     raise response_error
webapp-1  | minio.error.S3Error: S3 operation failed; code: XMinioStorageFull, message: Storage backend has reached its minimum free drive threshold. Please delete a few objects to proceed., resource: /pes2ug23cs100/image.jpg, request_
id: 189DE6A81F34E78F, host_id: dd9025bab4ad464b049177c95eb6ebf374d3b3fd1af9251148b658df7ac2e3e8, bucket_name: pes2ug23cs100, object_name: image.jpg
webapp-1  | 172.19.0.1 - - [18/Mar/2026 09:50:38] "POST /product HTTP/1.1" 500 -
```

Experiment 4:

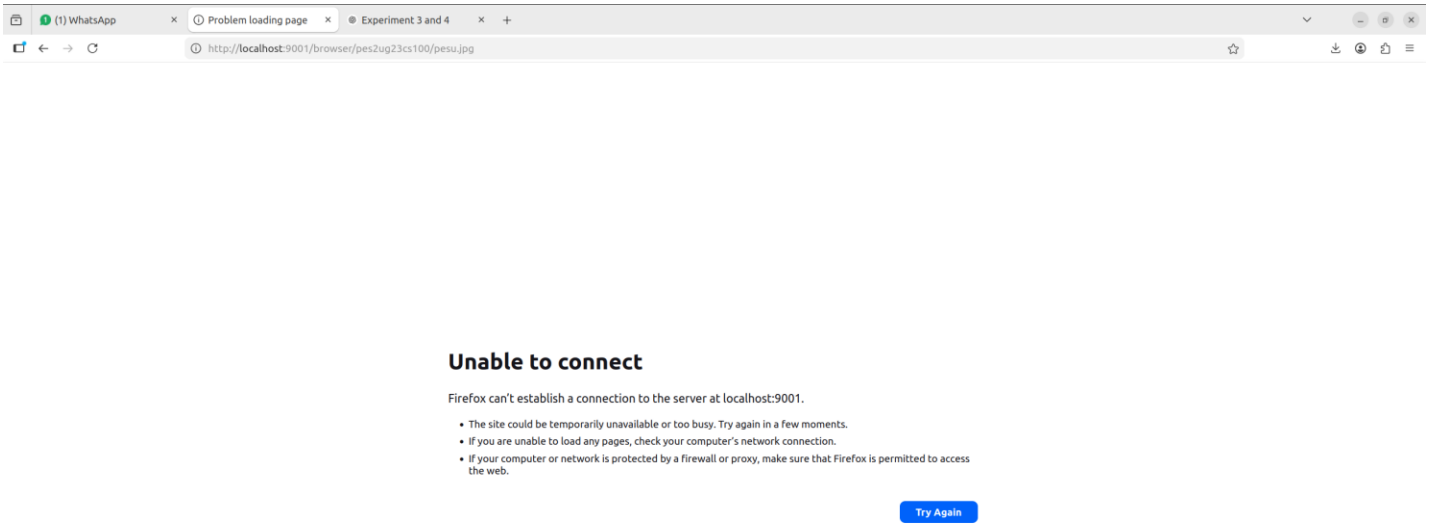
Minio is down (can’t upload image)

```
pes2ug23cs100@pes2ug23cs100: ~/CC Lab/Lab7/PES2UG23CS100          pes2ug23cs100@pes2ug23cs100: ~/CC Lab/Lab7/PES2UG23CS100
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ docker-compose stop minio
WARN[0000] /home/pes2ug23cs100/CC Lab/Lab7/PES2UG23CS100/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] stop 1/1
 ✔ Container pes2ug23cs100-minio-1 Stopped
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$
```

```
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ docker ps
CONTAINER ID   IMAGE                   COMMAND           CREATED          STATUS          PORTS                                          NAMES
6e3b83a58b75   pes2ug23cs100-webapp    "python app.py"   33 minutes ago   Up 33 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp    pes2ug23cs100-webapp-1
```

```
pes2ug23cs100@pes2ug23cs100:~/Pictures$ curl -X POST -F "name=Phone" -F "price=500" -F "image=@cricket.jpg" http://localhost:5000/product
<!doctype html>
<html lang=en>
<title>500 Internal Server Error</title>
<h1>Internal Server Error</h1>
<p>The server encountered an internal error and was unable to complete your request. Either the server is overloaded or there is an error in the application.</p>
pes2ug23cs100@pes2ug23cs100:~/Pictures$
```

```
webapp-1  | The above exception was the direct cause of the following exception:
webapp-1  |
webapp-1  | Traceback (most recent call last):
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 1511, in wsgi_app
webapp-1  |     response = self.full_dispatch_request()
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 919, in full_dispatch_request
webapp-1  |     rv = self.handle_user_exception(e)
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 917, in full_dispatch_request
webapp-1  |     rv = self.dispatch_request()
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 902, in dispatch_request
webapp-1  |     return self.ensure_sync(self.view_functions[rule.endpoint])(**view_args)  # type: ignore[no-any-return]
webapp-1  |   File "/app/app.py", line 67, in add_product
webapp-1  |     if not minio_client.bucket_exists(BUCKET_NAME):
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/minio/api.py", line 700, in bucket_exists
webapp-1  |     self._execute("HEAD", bucket_name)
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/minio/api.py", line 444, in _execute
webapp-1  |     return self._url_open(
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/minio/api.py", line 306, in _url_open
webapp-1  |     response = self._http.urlopen(
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/urllib3/poolmanager.py", line 457, in urlopen
webapp-1  |     response = conn.urlopen(method, u.request_uri, **kw)
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/urllib3/connectionpool.py", line 871, in urlopen
webapp-1  |     return self.urlopen(
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/urllib3/connectionpool.py", line 871, in urlopen
webapp-1  |     return self.urlopen(
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/urllib3/connectionpool.py", line 871, in urlopen
webapp-1  |     return self.urlopen(
webapp-1  |   [Previous line repeated 2 more times]
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/urllib3/connectionpool.py", line 841, in urlopen
webapp-1  |     retries = retries.increment(
webapp-1  |   File "/usr/local/lib/python3.9/site-packages/urllib3/util/retry.py", line 535, in increment
webapp-1  |     raise MaxRetryError(_pool, url, reason) from reason  # type: ignore[arg-type]
webapp-1  | urllib3.exceptions.MaxRetryError: HTTPConnectionPool(host='minio', port=9000): Max retries exceeded with url: /pes2ug23cs100 (Caused by NameResolutionError("HTTPConnection(host='minio', port=9000): Failed to resolve 'minio'
 ([Errno -3] Temporary failure in name resolution)"))
webapp-1  | 172.19.0.1 - - [18/Mar/2026 10:02:56] "POST /product HTTP/1.1" 500 -
```



Webapp is down (can see the db, but can't access it)

```
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ docker-compose start minio
WARN[0000] /home/pes2ug23cs100/CC Lab/Lab7/PES2UG23CS100/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] start 1/1
 ✔ Container pes2ug23cs100-minio-1 Started
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ docker-compose stop webapp
WARN[0000] /home/pes2ug23cs100/CC Lab/Lab7/PES2UG23CS100/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] stop 1/1
 ✔ Container pes2ug23cs100-webapp-1 Stopped
```

```
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                                                                   NAMES
f6160a31bf35   minio/minio    "/usr/bin/docker-ent…"   38 minutes ago   Up 52 seconds   0.0.0.0:9000-9001->9000-9001/tcp, [::]:9000-9001->9000-9001/tcp   pes2ug23cs100-minio-1
```

```
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ ls my_block_data
ecommerce.db
```

```
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ docker exec -it pes2ug23cs100-webapp-1 python -c "import sqlite3;
conn = sqlite3.connect('/mnt/block_volume/ecommerce.db'); cursor = conn.cursor(); cursor.execute('SELECT * FROM product');
print(cursor.fetchall()); conn.close()"
Error response from daemon: container 6e3b83a58b75865e014d2ca9fdbd689b57bf8a01231600fe0dba6202ca607c5d is not running
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$
```

```
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ docker-compose start webapp
WARN[0000] /home/pes2ug23cs100/CC Lab/Lab7/PES2UG23CS100/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] start 1/1
 ✔ Container pes2ug23cs100-webapp-1 Started                                                                    0.5s
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS          PORTS                                                                                   NAMES
6e3b83a58b75   pes2ug23cs100-webapp   "python app.py"          43 minutes ago   Up 4 seconds    0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp                                             pes2ug23cs100-webapp-1
f6160a31bf35   minio/minio            "/usr/bin/docker-ent…"   43 minutes ago   Up 5 minutes    0.0.0.0:9000-9001->9000-9001/tcp, [::]:9000-9001->9000-9001/tcp   pes2ug23cs100-minio-1
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$ docker exec -it pes2ug23cs100-webapp-1 python -c "import sqlite3; conn = sqlite3.connect('/mnt/block_volume/ecommerce.db'); cursor = conn.cursor(); cursor.execute('SELECT * FROM
product'); print(cursor.fetchall()); conn.close()"
[(1, 'Phone', 500.0, 'cat.jpg'), (2, 'Phone', 500.0, 'pesu.jpg'), (3, 'Phone', 500.0, 'pesu.jpg'), (4, 'Phone', 500.0, 'pesu.jpg'), (5, 'Phone', 500.0, 'cat.jpg'), (6, 'Phone', 500.0, 'test.jpg')]
pes2ug23cs100@pes2ug23cs100:~/CC Lab/Lab7/PES2UG23CS100$
```